Index page

Page 2 Block Diagram

Page 3 GD30BM2016 and some of its peripheral circuits、Multi-cell battery interface

Page 4 Thermistor Temperature Measurement
Multifunction pins (TS1, TS2, TS3, CFETOFF, DFETOFF, ALERT, HDQ, DCHG, and DDSG)

Page 5 Cell Balance (NMOS)

Page 6 Three Parallel FET Test Circuits

Page 7 Communication circuits

Page 8 Interface and GD32E503CCT6 Microcontroller

Page 9 Mounting holes, Fiducials Marking

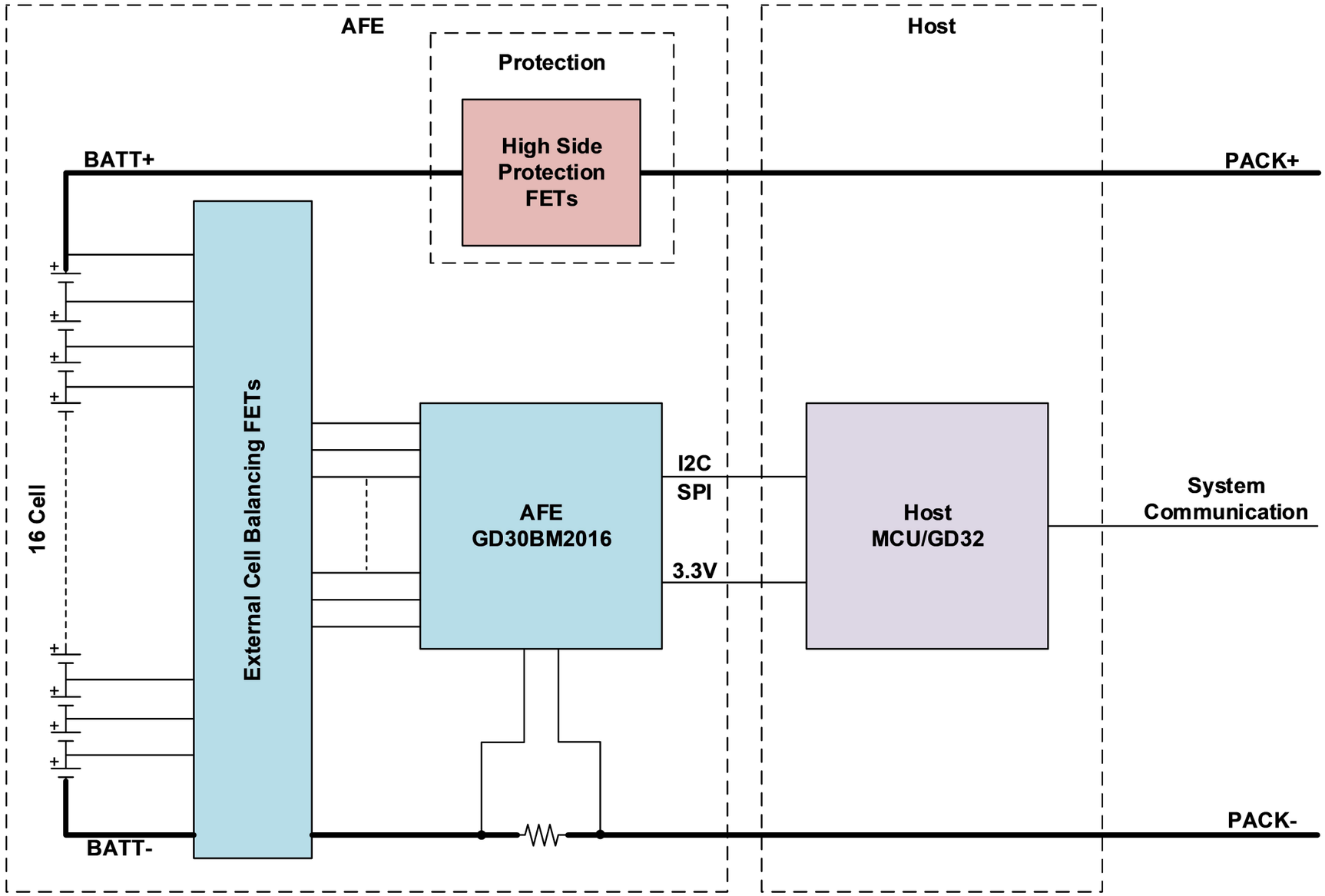
Revision History

Revision	Notes
A	Initial
B	SCH 1. Add J1 between BAT+ and C16, Add J7 beteween BAT- and C0 2. Add D4/D5 between VC0 and VSS 3. Add D6 beteween VC1 and VSS 4. Connect the TS2 pin to the MCU PB11 pin 5. Connect the RST_SHUT pin to the MCU PB12 pin 6. Remove cell simulator circuits 7. Add additional configurable pins for PA1, PA2 and PA3 PCB 1. Optimize the routing of VC0 and VC16

GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19
Number: BM001	Project Title: GD30BM2016 EVB	
Rev: B	Sheet Title: Cover	
SVN Rev: NA	Assembly Variant: 301	Sheet: 1 of 9
Drawn By: yiwen.ren	File: Cover.SchDoc	Size: A3
Engineer: yiwen.ren	Contact: NA	

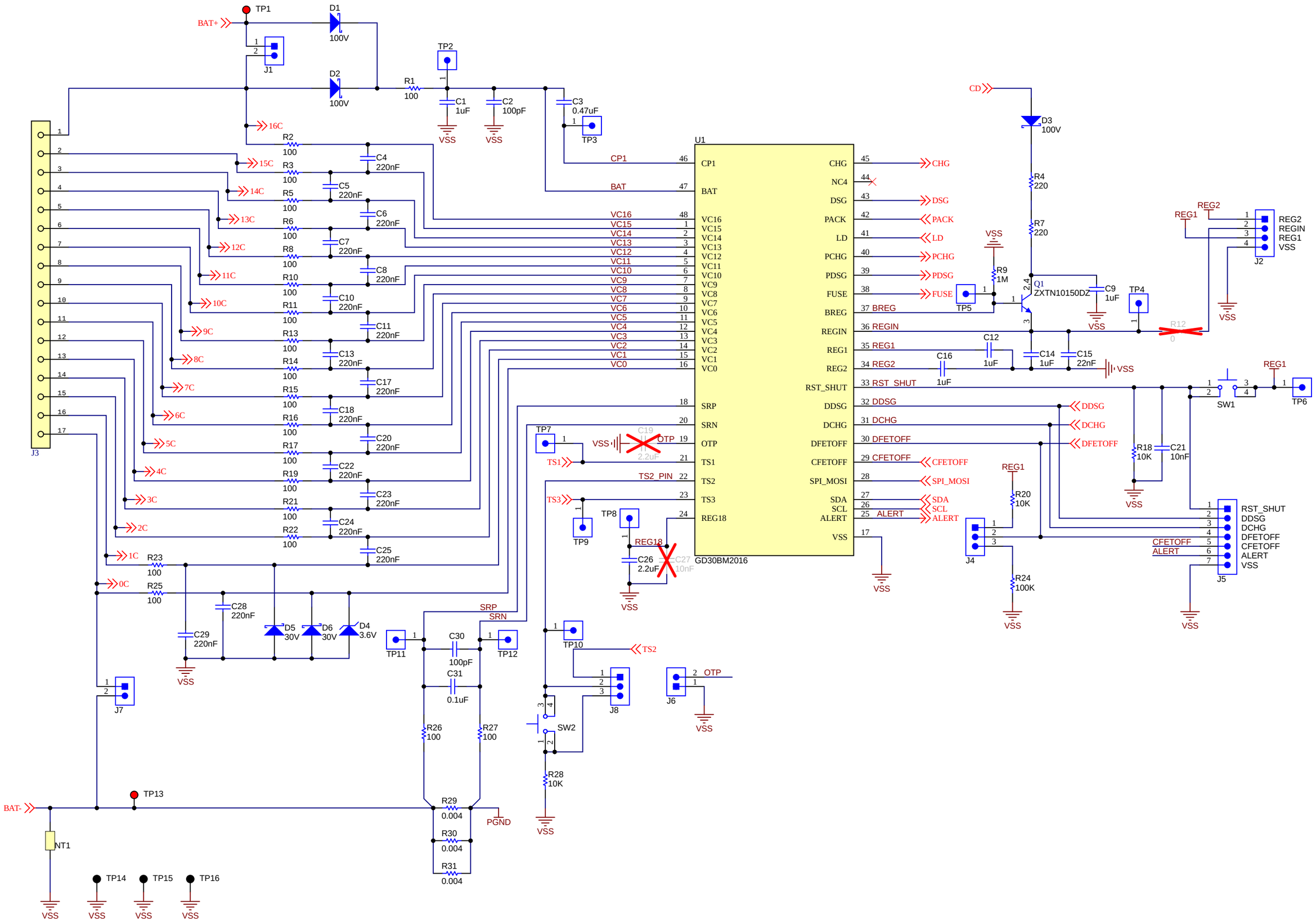
GigaDevice
http://www.gigadevice.com
© GigaDevice 2025



GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/5/29
Number: BM001	Project Title: GD30BM2016 EVB	
Rev: B	Sheet Title: BD	
SVN Rev: NA	Assembly Variant: 301	Sheet: 2 of 9
Drawn By: yiwen.ren	File: BlockDiagram.SchDoc	Size: A3
Engineer: yiwen.ren	Contact: NA	

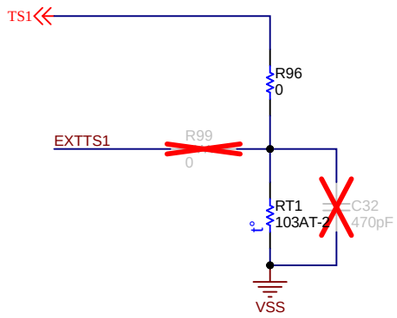
GigaDevice
http://www.gigadevice.com
© GigaDevice 2025

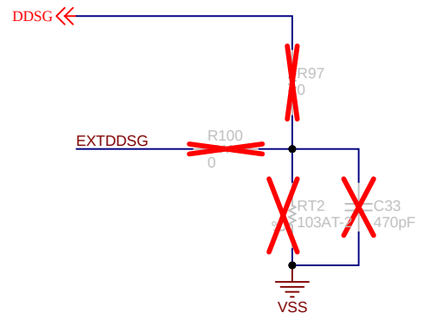


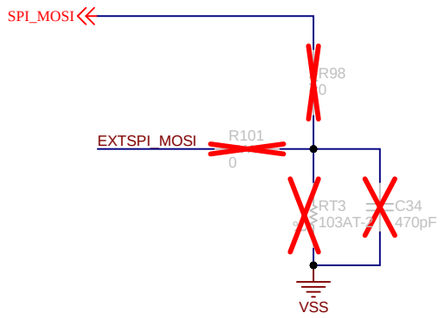
GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19
Number: BM001	Project Title: GD30BM2016 EVB	
Rev: B	Sheet Title: AFE Monitor	
SVN Rev: NA	Assembly Variant: 301	Sheet: 3 of 9
Drawn By: yiwen.ren	File: BM001B_AFE.SchDoc	Size: A3
Engineer: yiwen.ren	Contact: NA	

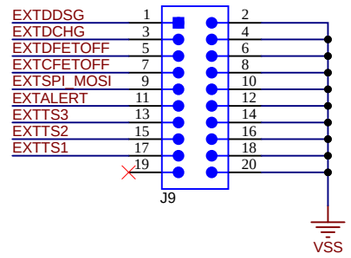
GigaDevice
http://www.gigadevice.com
© GigaDevice 2025

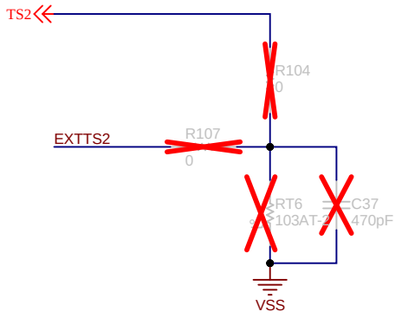


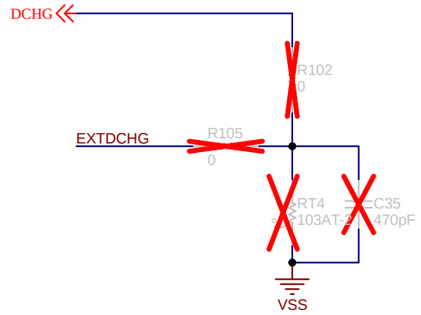


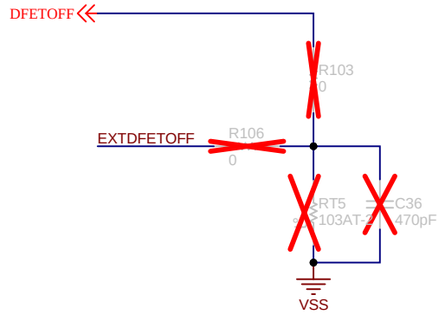


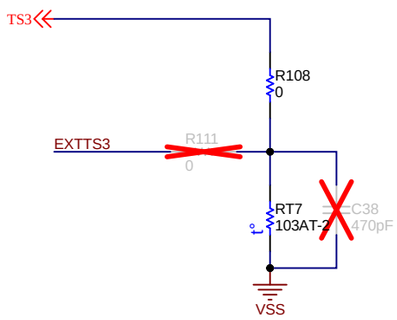
External thermistor interface

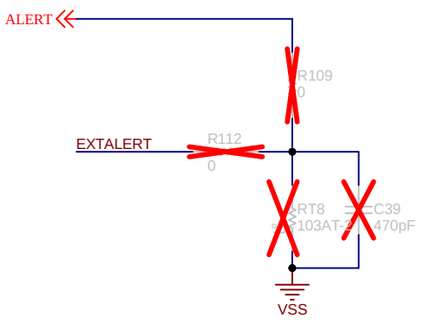


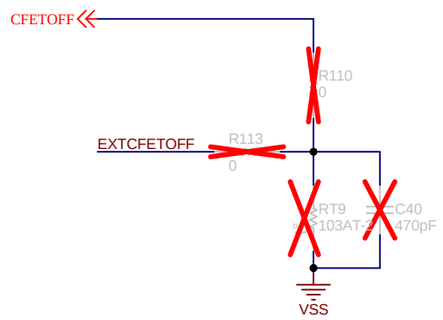








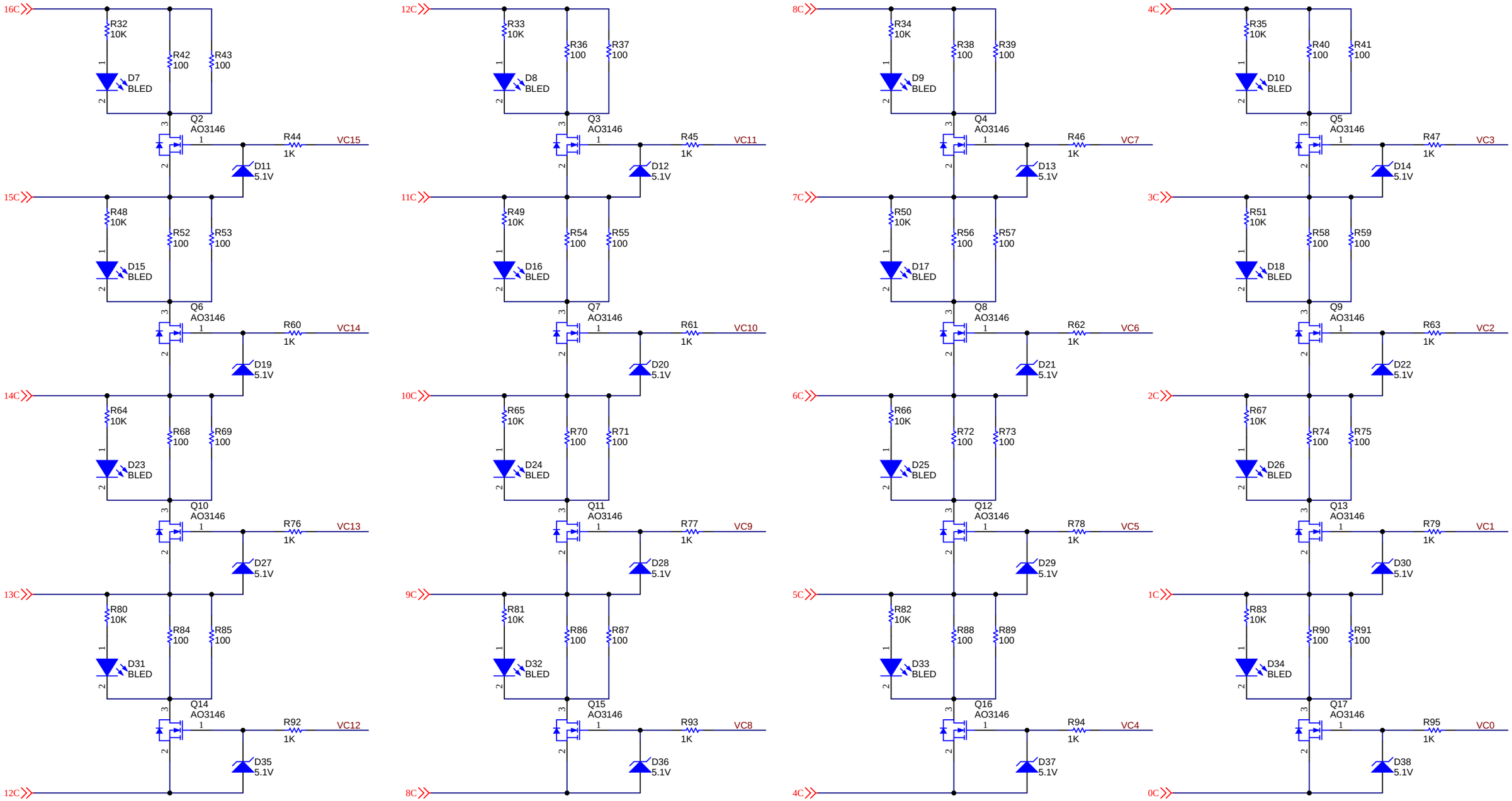




GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19	GigaDevice
Number: BM001	Project Title: GD30BM2016 EVB		http://www.gigadevice.com
Rev: B	Sheet Title: TermistorTemp Measurement		
SVN Rev: NA	Assembly Variant: 301	Sheet: 4 of 9	
Drawn By: yiwen.ren	File: BM001B_TermistorTemp.SchDoc	Size: A3	
Engineer: yiwen.ren	Contact: NA		

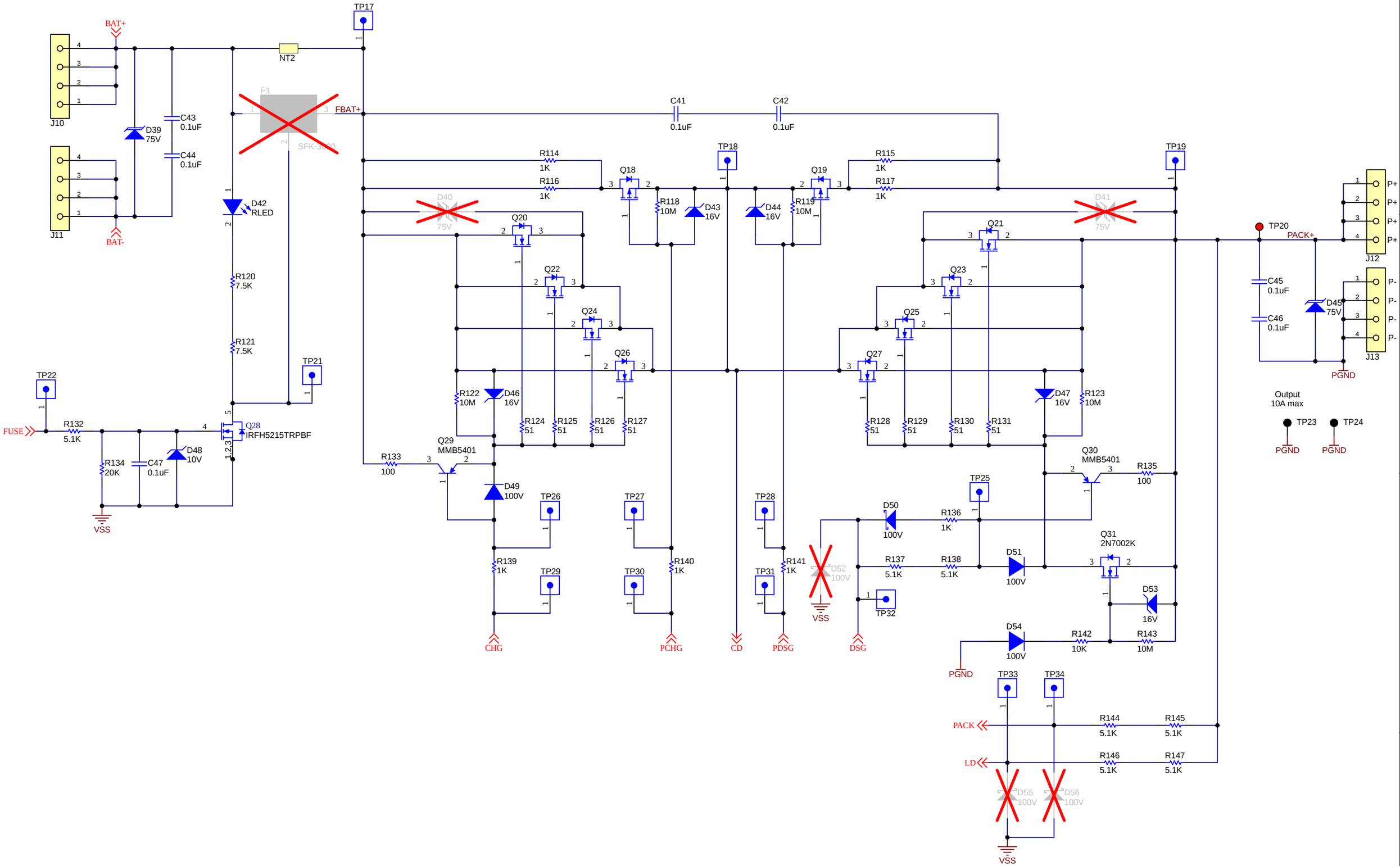
© GigaDevice 2025



GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19	
Number: BM001	Project Title: GD30BM2016 EVB		
Rev: B	Sheet Title: Cell Balance		
SVN Rev: NA	Assembly Variant: 301	Sheet: 5 of 9	
Drawn By: yiwen.ren	File: BM001B_Cell Balance.SchDoc	Size: A3	
Engineer: yiwen.ren	Contact: NA		

GigaDevice
http://www.gigadevice.com
© GigaDevice 2025



GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19
Number: BM001	Project Title: GD30BM2016 EVB	
Rev: B	Sheet Title: Multiple FETs	
SVN Rev: NA	Assembly Variant: 301	Sheet: 6 of 9
Drawn By: yiwen.ren	File: BM001B_Multiple FETs.SchDoc	Size: A3
Engineer: yiwen.ren	Contact: NA	

GigaDevice
http://www.gigadevice.com
© GigaDevice 2025

Multi-function Pins for External Communication

CFETOFF / SPI_CS
SPI_MOSI
SDA / SPI_MISO
SCL / SPI_SCLK

Pull-up / pull-down selection

GD32 MCU interface type selection

TP35 REG1 CFETOFF R148 10K R149 100K J15 J14 GD32_SPI_CS EXT_SPI_CS VSS

TP36 REG1 SPI_MOSI R150 10K R151 100K J17 J16 GD32_SPI_MOSI EXT_SPI_MOSI VSS

TP37 REG1 SDA R152 10K J18 J19 GD32_SPI_MISO_A1 GD32_IIC_SDA EXT_SPI_MISO

TP38 REG1 SCL R153 10K J20 J21 GD32_SPI_SCK GD32_IIC_SCL EXT_SPI_CLK

External I2C Connection

External SPI Connections

J22
1 SPI_MOSI
2 SPI_MISO SDA
3 SPI_SCK SCL
4 SPI_CS
5 PACK-

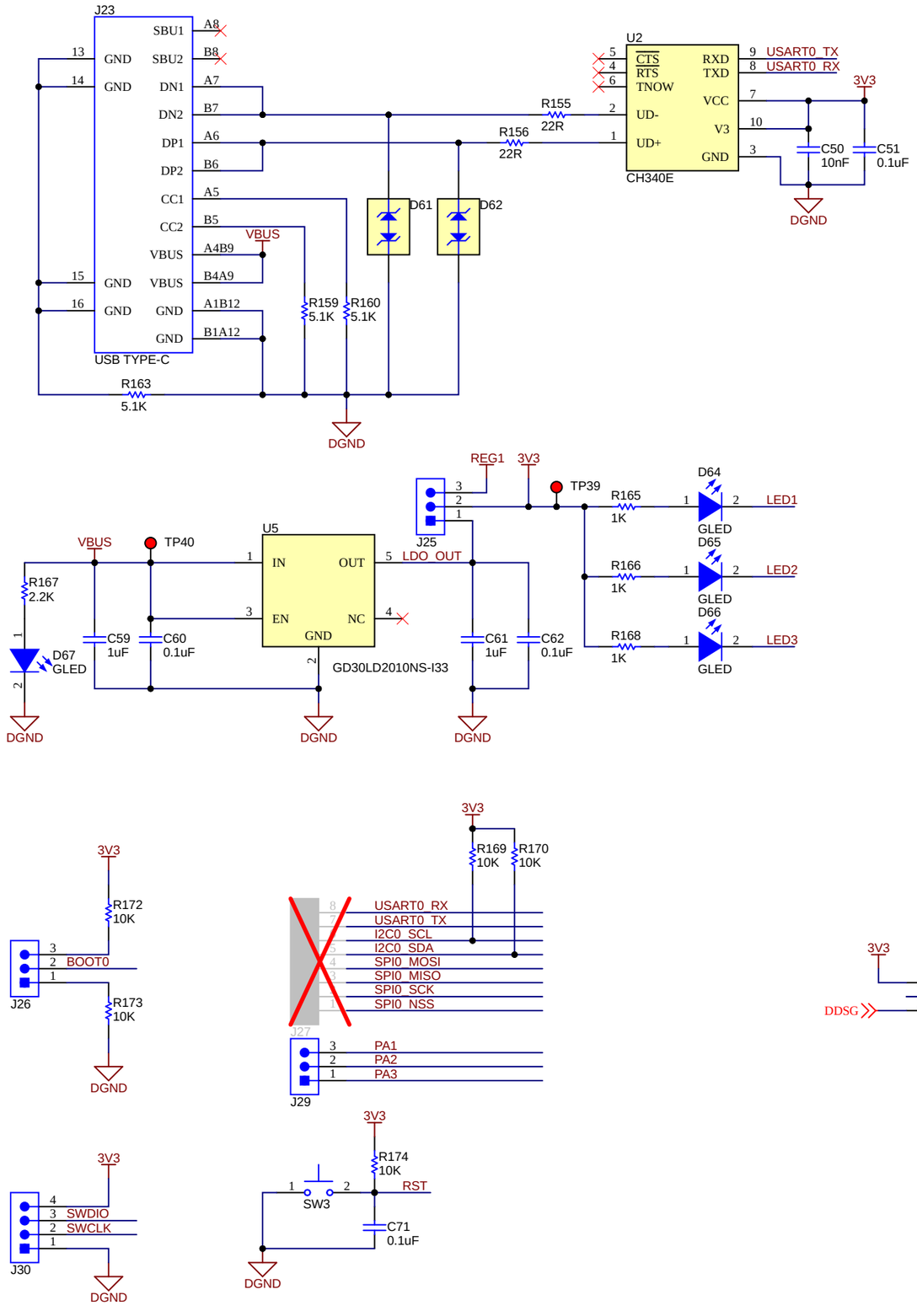
D57 D58 D59 D60 E1 E2 E3 E4

PGND

GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19
Number: BM001	Project Title: GD30BM2016 EVB	
Rev: B	Sheet Title: CommConfig	
SVN Rev: NA	Assembly Variant: 301	Sheet: 7 of 9
Drawn By: yiwen.ren	File: BM001B_CommConfig.SchDoc	Size: A3
Engineer: yiwen.ren	Contact: NA	

GigaDevice
http://www.gigadevice.com
© GigaDevice 2025

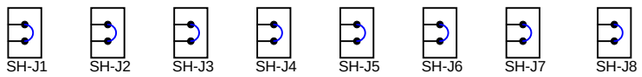


GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/19
Number: BM001	Project Title: GD30BM2016 EVB	
Rev: B	Sheet Title: Interface	
SVN Rev: NA	Assembly Variant: 301	Sheet: 8 of 9
Drawn By: yiwen.ren	File: BM001B_Interface.SchDoc	Size: A3
Engineer: yiwen.ren	Contact: NA	

GigaDevice
http://www.gigadevice.com
© GigaDevice 2025









PCB
LOGO
GigaDevice LOGO

PCB
LOGO
WEEE LOGO

PCB Number: BM001
PCB Rev: B

Variant/Label Table	
Variant	Label Text
301	ChangeMe!

ZZ1
Label Assembly Note
This Assembly Note is for PCB labels only.

ZZ2
Assembly Note
These assemblies are ESD sensitive, ESD precautions shall be observed.

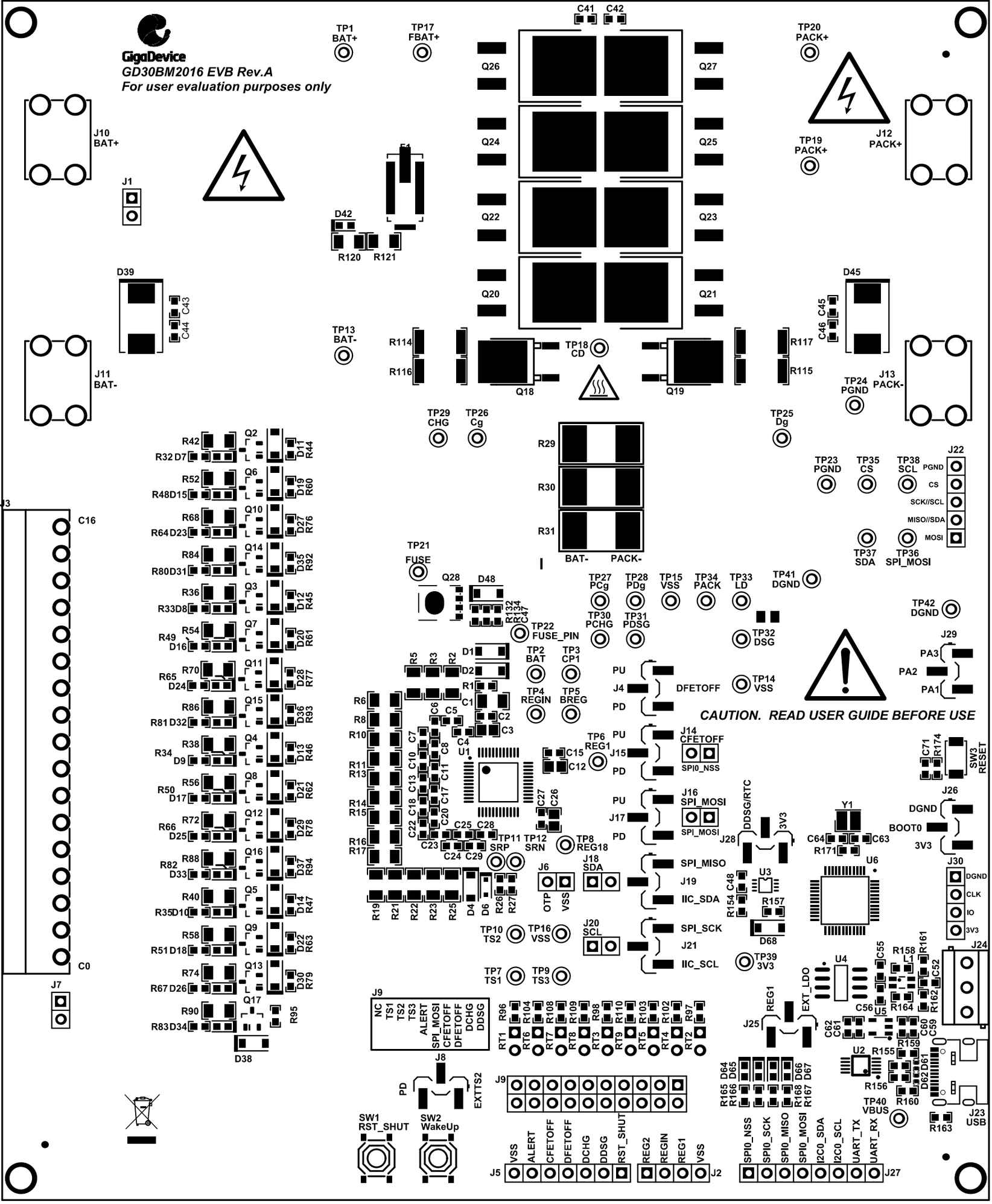
ZZ3
Assembly Note
These assemblies must comply with workmanship standards IPC-A-610 Class 2, unless otherwise specified.

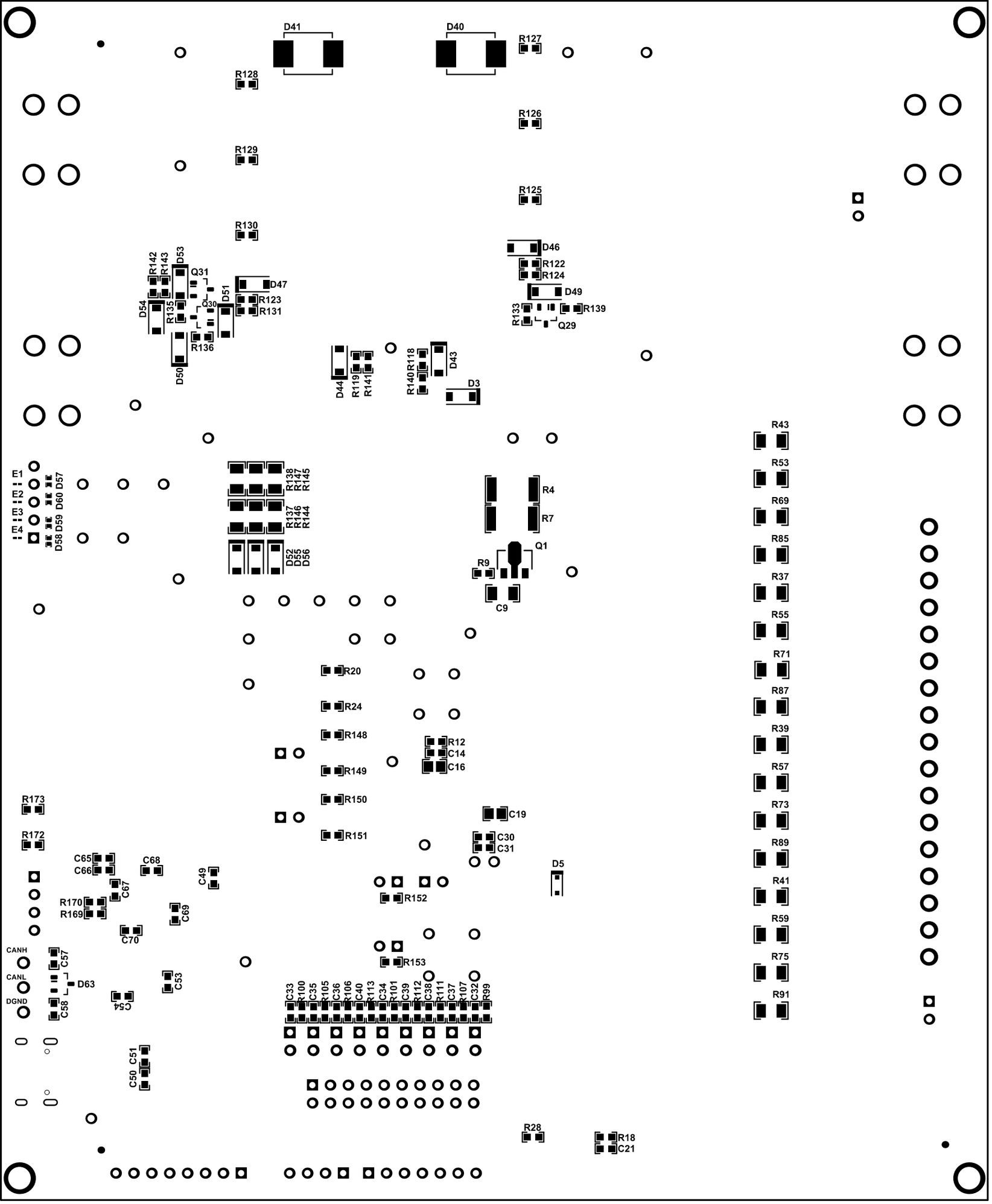
ZZ4
Assembly Note
These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.

GigaDevice and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. GigaDevice and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. GigaDevice and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

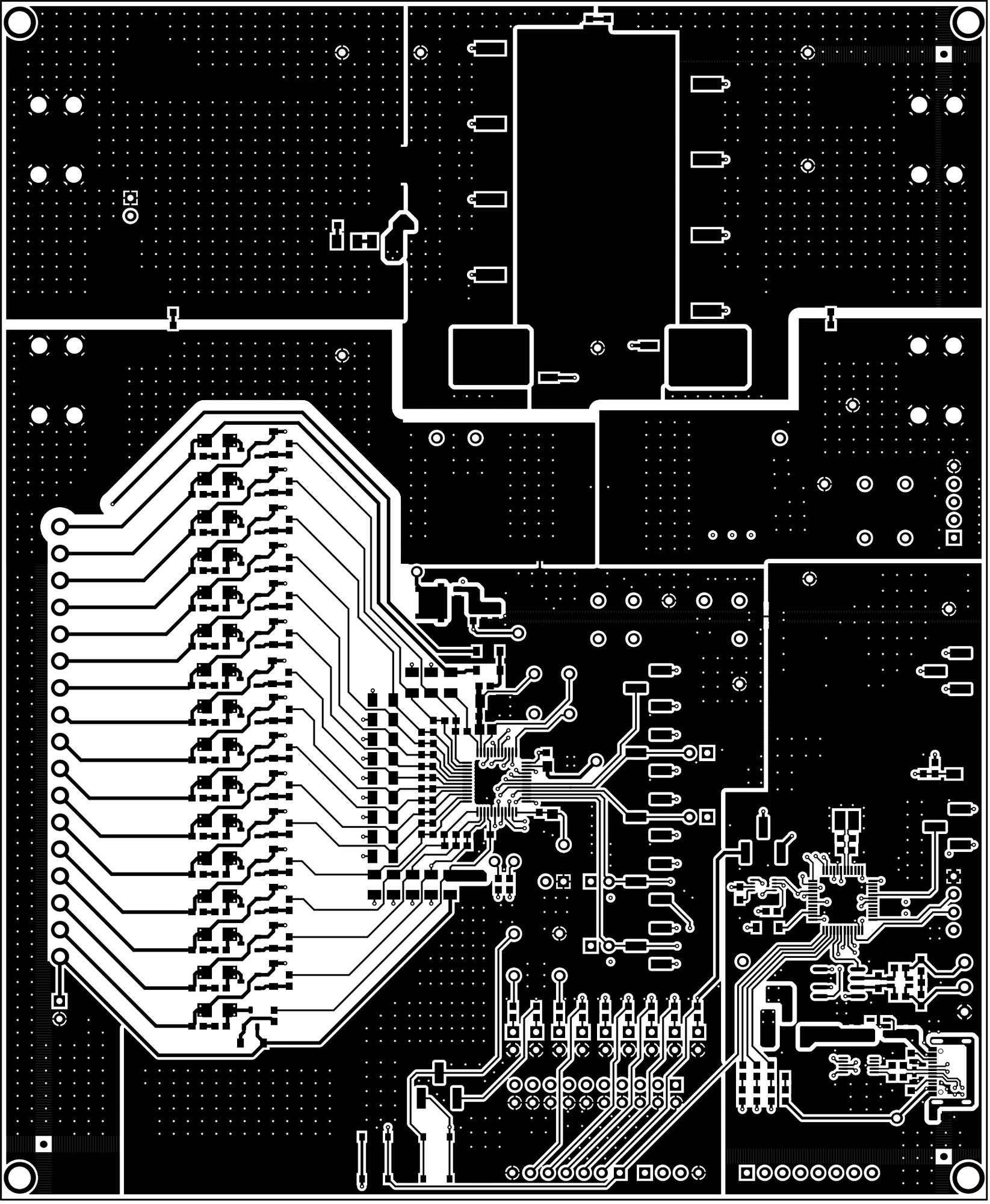
Orderable: NA	Designed for: Plublic Release	Mod. Date: 2025/9/15	
Number: BM001	Project Title: GD30BM2016 EVB		
Rev: B	Sheet Title: Hardware		
SVN Rev: NA	Assembly Variant: 301	Sheet: 9 of 9	
Drawn By: yiwen.ren	File: BM001B_Hardware.SchDoc	Size: A3	
Engineer: yiwen.ren	Contact: NA		

GigaDevice
http://www.gigadevice.com
© GigaDevice 2025

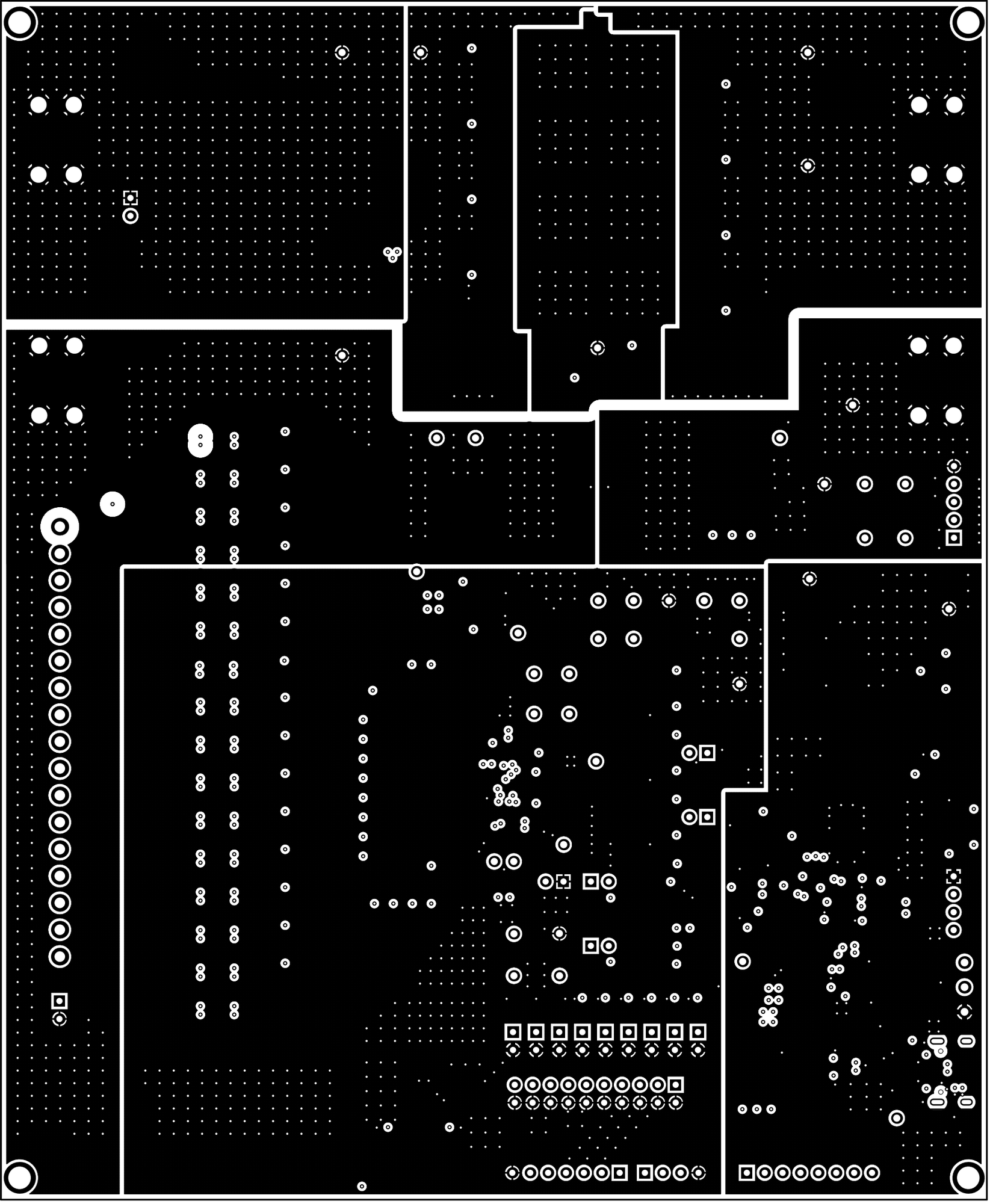

GigaDevice
GD30BM2016 EVB Rev.A
For user evaluation purposes only
CAUTION. READ USER GUIDE BEFORE USE
Top Overlay

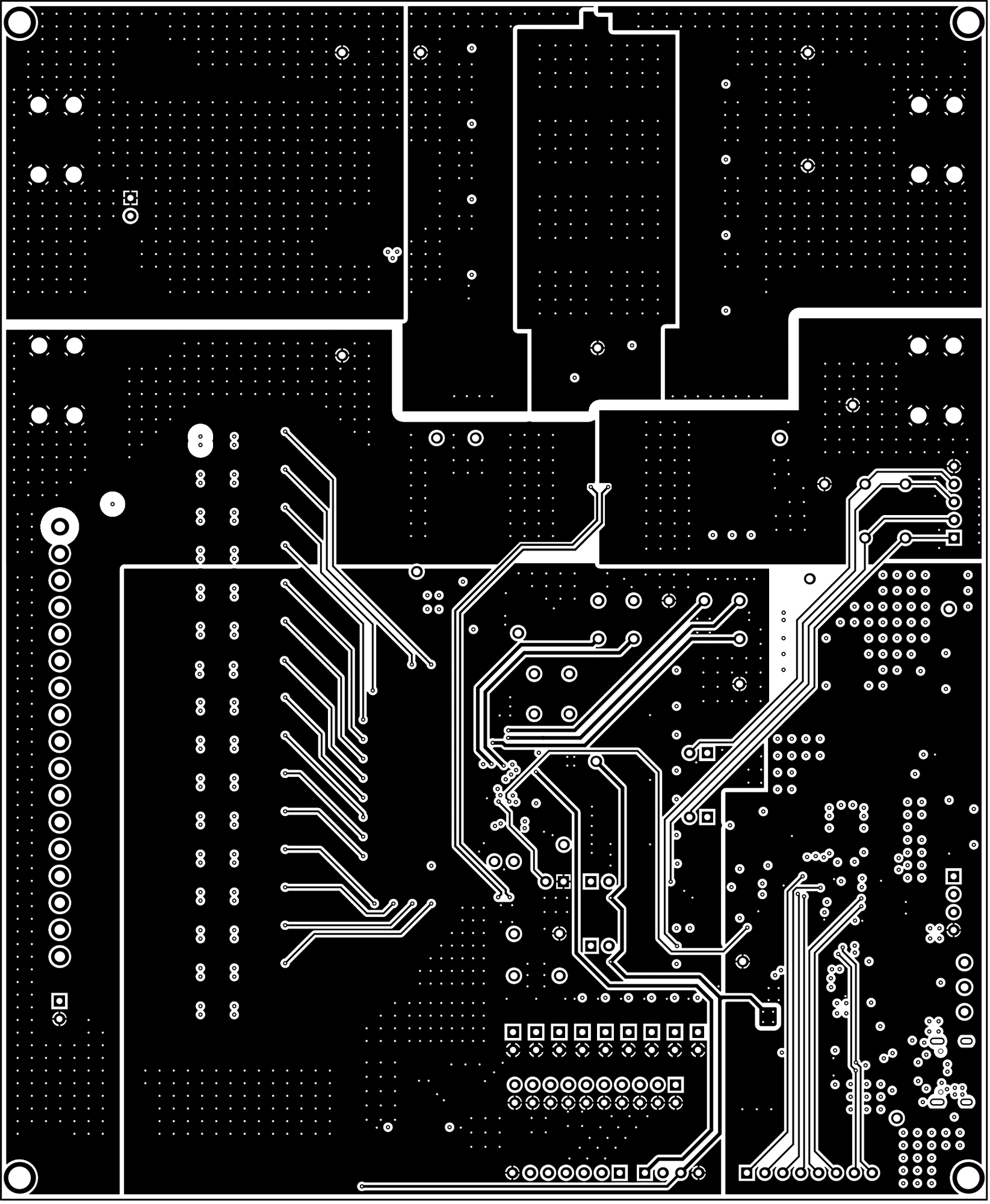


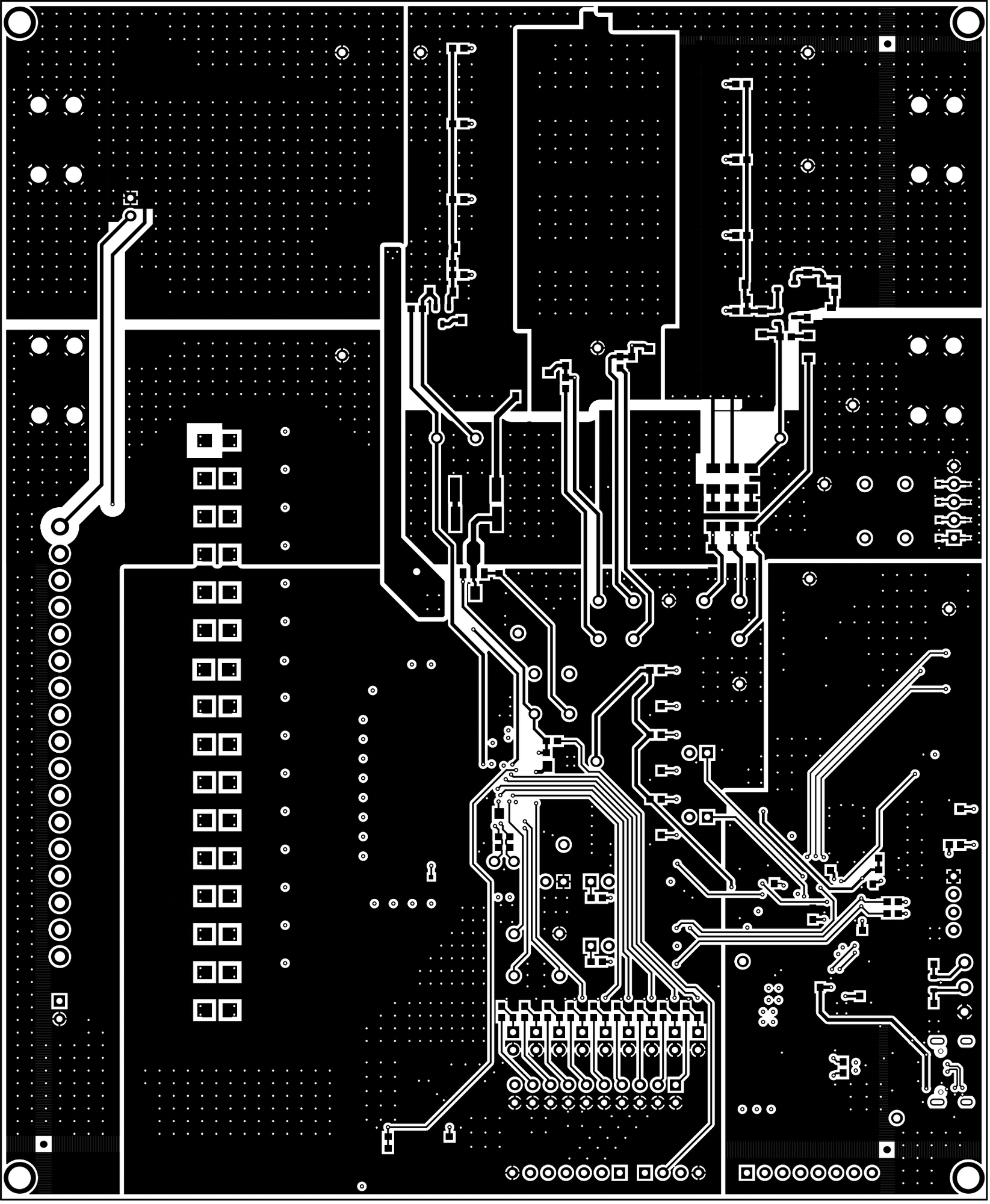
Bottom Overlay



TOP

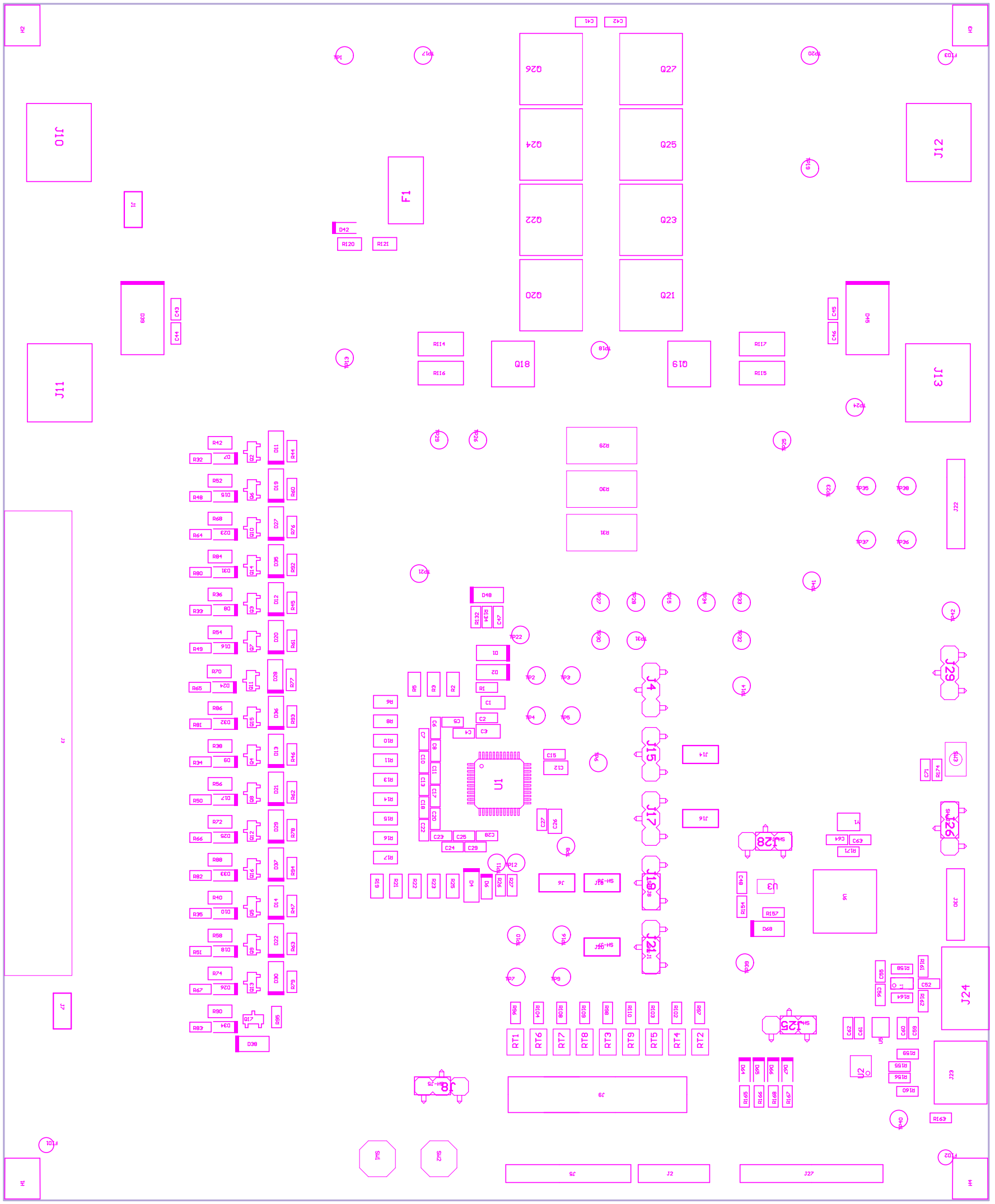






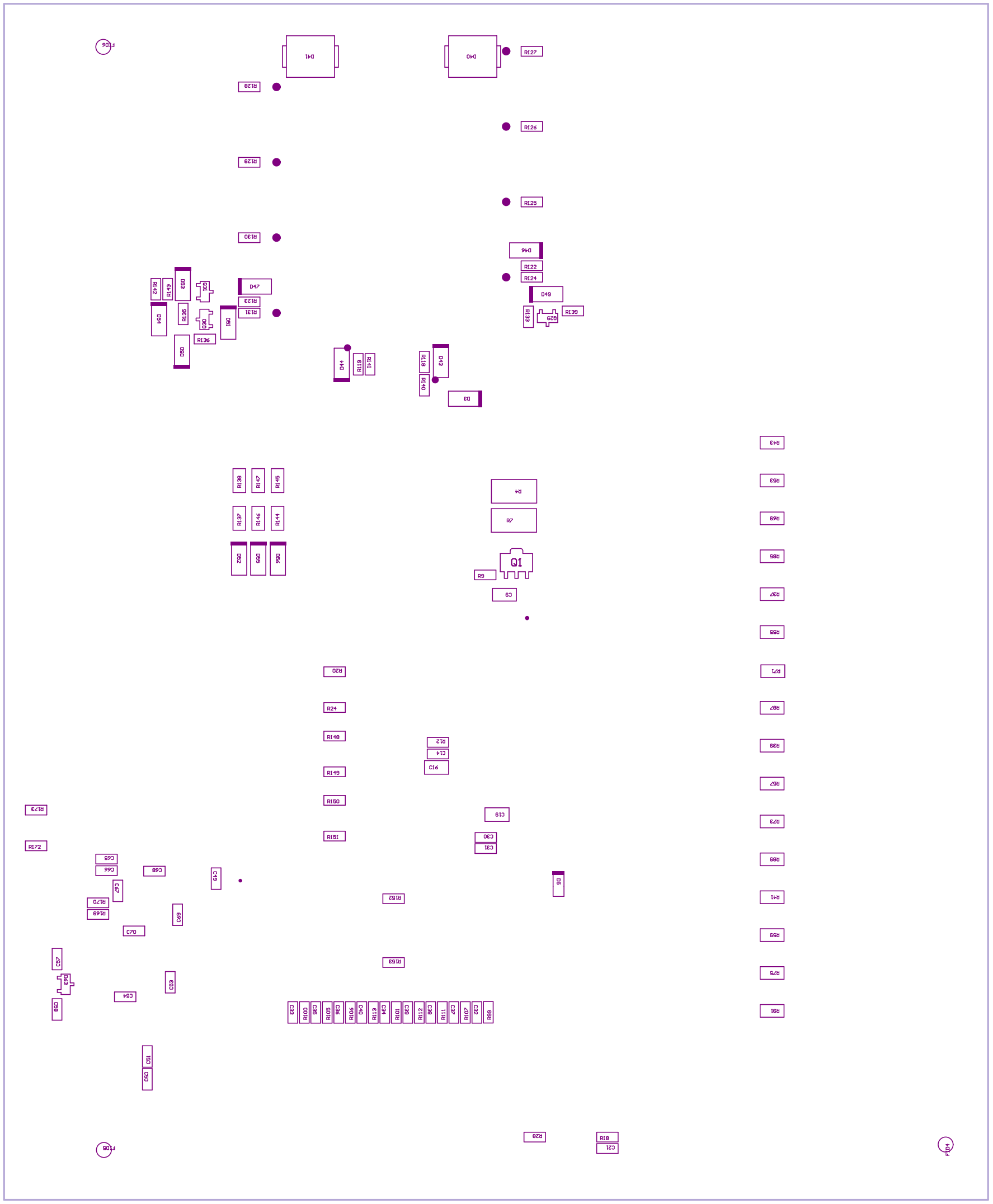
BOTTOM

ZZ1 ■ Install label in silkscreened box after final wash. Text shall be truetype font. Text shall be per the Label Table in the PDF schematic.
ZZ2 ■ These assemblies are ESD sensitive, ESD precautions shall be observed.
ZZ3 ■ These assemblies must comply with workmanship standards IPC-A-610 Class 2, unless otherwise specified.
ZZ4 ■ These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.



COMPONENTS MARKED 'DNP' SHOULD NOT BE POPULATED.
ASSEMBLY VARIANT: [No Variations]

PCB VIEWED FROM TOP SIDE	BOARD #: BM001	SVN REV: NA
LAYER NAME = M5 Assembly Top	REV: B	Sheet Size: A3
PLOT NAME = GD30BM2016 EVB	GENERATED : 2025/9/19 14:50:12	GIGA DEVICE



PCB VIEWED FROM TOP SIDE	BOARD #: BM001	SVN REV: NA
	REV: B	M6 Assembly Bottom
PLOT NAME = GD30BM2016 EVB	GENERATED : 2025/9/19 14:50:12	GIGA DEVICE